7.9 Variable Reluctance Tachogenerator

It can be used for measurement of linear and angular velocity. Magnetomotive force (mmf) is the force that causes flux to be established and it is analogous to the electromotive force for electric circuits. The unit of mmf in SI units is the Ampere, but it refers to a coil of one turn.

The opposition to the establishment of magnetic flux is called *reluctance*. Mathematically, reluctance is defined as

$$\Re = \frac{\text{mmf}}{\phi}$$

Rearranging we get

$$\text{mmf} = \Re \times \phi$$

where mmf is in Ampere turns and ϕ (flux) is in Weber.

7.9.1 *Principle of Operation*

If the time-varying flux ϕ is linked by a single turn of coil, then the back emf developed in the coil can be expressed as

$$e = -\frac{d\phi}{dt}$$

A variable reluctance tachogenerator is shown in Fig. 7.22. It consists of toothed wheel that is made of ferromagnetic material and a coil wound on a permanent magnet.

The permanent magnet is extended by a soft iron pole piece. The teeth of the wheel move in close proximity to the pole piece. Therefore, the flux linked by the coil changes with time and voltage is developed across the coil. The total flux (ϕ_T) linked by a coil of m turn is given by

$$\phi_T = m\phi = m\frac{\text{mmf}}{\Re}$$

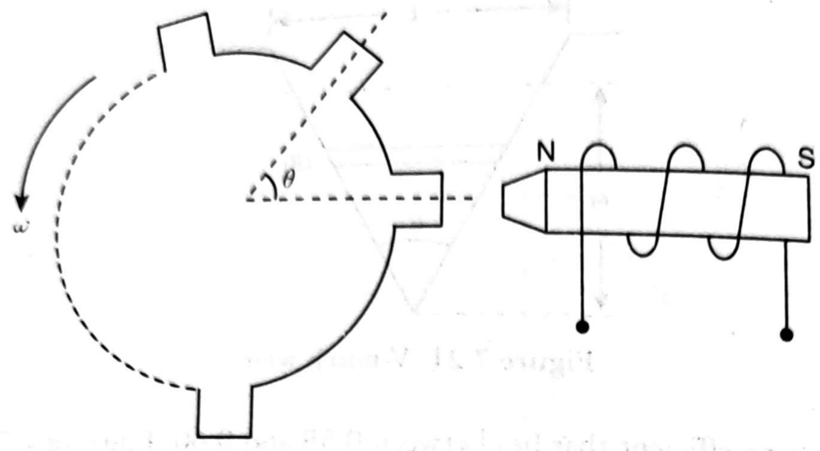

Figure 7.22 A variable reluctance tachogenerator.

Moreover, we know when the reluctance is minimum, the flux is maximum and vice versa. The variation of flux (ϕ_T) with the angular position (θ) of the wheel can be expressed as

$$\phi_T(\theta) = \alpha + \beta \cos(n\theta)$$

where α is the mean flux, β is the amplitude of the time-varying flux, and n is the number of teeth of the wheel. The induced emf can be expressed by

$$e = -\frac{d\phi_T}{dt} = -\frac{d\phi_T}{d\theta}\frac{d\theta}{dt}$$

Again,

$$\frac{d\phi_T}{d\theta} = -\beta n \sin(n\theta)$$

and

$$\theta = \omega t$$

Therefore

$$\frac{d\theta}{dt} = \omega$$

where ω is the rotational velocity of the wheel. Therefore,

$$e = \beta n\, \omega \sin n\omega t$$

7.10 Turbine Flowmeter

It has direct electrical output. This flowmeter consists of a wheel with multiple blades and it is installed inside the pipe in which clean fluid is flowing. A schematic of turbine flowmeter is shown in Fig. 7.23.

The turbine meter must be installed along an axis parallel to the direction of fluid flow in the pipe. The flow of the fluid past the wheel causes it to rotate at a rate proportional to volume flow rate. The wheel or blade of the meter is shown in Fig. 7.24.

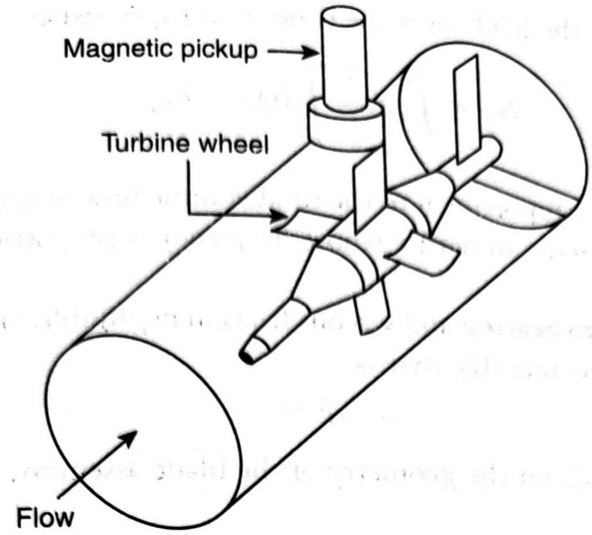

Figure 7.23 Schematic of turbine flow meter.

7.10.1 Construction

The construction of a turbine flowmeter is similar to that of the variable reluctance tachogenerator. The blades or teeth are made of ferromagnetic material. The magnetic pickup consists of a coil wound on a permanent magnet. A voltage pulse is obtained at the output of the pickup whenever a turbine teeth or blade passes the pickup coil. The flow rate can be determined by measuring the pulses by a pulse counter.

7.10.2 Theory

If the number of rotation of blade per second is 'f' and the volume flow rate is 'Q', then $f = kQ$, where k is the sensitivity of the flowmeter. So 'Q' can be determined by measuring 'f'. The total volume of the liquid (V_T) flown through the pipe over a given time 'T' can be expressed as

$$V_T = \int_0^T Q\,dt$$

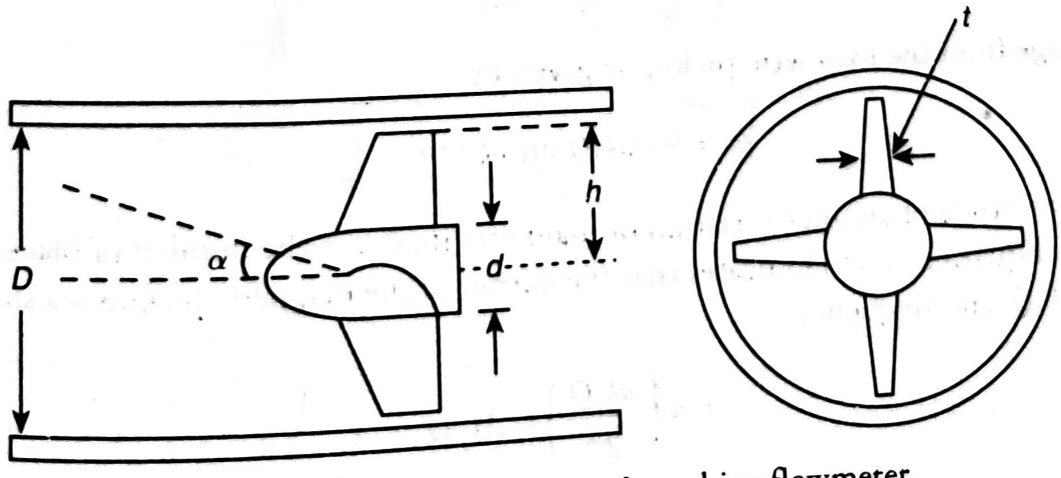

Figure 7.24 Blade assembly of a turbine flowmeter.

The total number of rotation of the blade over the time 'T' is expressed as

$$N_T = \int_0^T f dt = \int_0^T kQ dt = kV_T$$

Therefore the total count will be proportional to the total volume flow of liquid. The voltage induced in the magnetic pickup will be periodic in nature whose frequency is proportional to the angular velocity of blades.

Assuming the drag force due to bearing and viscous friction negligible, the rotor angular velocity (ω) is proportional to the volume flow rate (Q), that is,

$$\omega = k_1 Q$$

where k_1 is a constant that depends on the geometry of the blade assembly. The velocity of the fluid 'u' is given by

$$u = \frac{Q}{A}$$

where

A = Area of cross-section of the pipe – Area of the cross-section of hub – Area of the blades

Mathematically,

$$A = \frac{\pi D^2}{4} - \frac{\pi d^2}{4} - n\left(h - \frac{d}{2}\right)t$$

where n is the number of blades; t is their average thickness; h and d are the dimensions of the blades system as shown in Fig. 7.24. Moreover,

$$\frac{\omega h}{u} = \tan\alpha$$

where ωh is the velocity of the blade tip perpendicular to the direction of flow of the fluid through the pipe and α is the inlet angle at the tip.

Again,

$$k_1 = \frac{\omega}{Q} = \frac{\tan\alpha}{Ah}$$

The output voltage from the magnetic pickup is given by

$$\boxed{e = \beta n k_1 Q \sin(n k_1 Q)t} \tag{7.16}$$

where β is the amplitude of angular variation of magnetic flux; n is the number of blades; and Q is the volume flow rate. Equation (7.16) indicates that the output of the magnetic pickup is a sinusoidal signal of amplitude '$\beta n k_1 Q$' and frequency

$$f = \left(\frac{n k_1 Q}{2\pi}\right) = M_f Q$$

Therefore, both the amplitude and frequency of the output signal vary with the flow rate. The flowmeter output signal is passed through an integrator and Schmitt trigger. Then the output will be a constant amplitude square wave of frequency proportional to flow rate.

7.10.3 Salient Features

If a turbine wheel is mounted in low friction bearings, the measurement accuracy can be as high as ±0.1%. It is less rugged and reliable than restriction type differential pressure flowmeter. The blades and bearings can be damaged if solid particles are present in the liquid. It is more expensive and it also imposes the permanent pressure loss on the measurement system. The typical range is usually 10% to 100% of full-scale output reading. Below 10% of full-scale reading, the bearing friction makes the meter reading inaccurate. Turbine meters are particularly prone to large errors when there is any significant second phase in the fluid being measured. Turbine meters have similar cost and market share to positive displacement meter. The former are smaller and lighter than the latter. It is preferred for low viscosity, high flow measurement.

7.11 Electromagnetic Flowmeter

Electromagnetic flowmeter is limited to measuring the volume flow rate of electrically conductive fluids (the conductivity of fluid should be greater than 10 μmho$\,$cm^{-1}). The schematic of an electromagnetic flowmeter is shown in Fig. 7.25. The instrument consists of a cylindrical tube made of stainless steel fitted with an insulating liner.

The typical lining materials are neoprene, polyterafluroethylene (PTFE), and polyurethane. A magnetic field that is perpendicular to the direction of flow of liquid is created in the tube by installing energised field coils in diametrically opposite side of the pipe. The voltage induced in the fluid is measured by two electrodes inserted into the opposite sides of the tube. The ends of the electrodes are flush with the inner surface of the fluid-carrying pipe. The materials used for making the electrodes are stainless steel, platinum iridium alloys, titanium, and tantalum. In accordance with the Faraday's law of

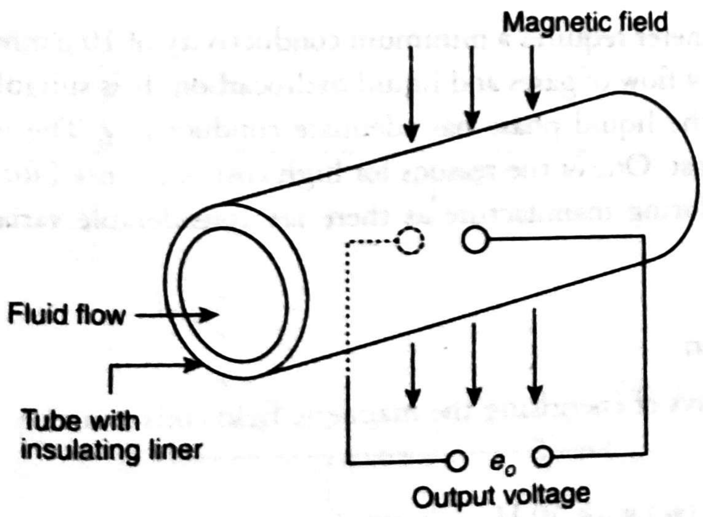

Figure 7.25 Electromagnetic flowmeter.

electromagnetic induction, the voltage 'e' induced across the length 'L' of the flowing fluid flowing at a velocity 'V' in a magnetic field of flux density B is given by

$$e = BLV \text{ Volt}$$

where B (flux density) is in Weber/(metre)^2 or Tesla; L (distance between the electrodes or inner diameter of the pipe) is in metre; V (velocity of the fluid) is in metre/sec.

7.11.1 Advantages

There is no obstruction of the fluid flow. Therefore, there is no pressure loss associated with the measurement. It is suitable for installation that can tolerate only a small pressure drop. The absence of any internal part is very attractive for measurement of velocity of corrosive and dirty fluids. The operating principle is independent of fluid density and viscosity and the output voltage is proportional to average velocity of fluid. There is no difficulty of measurement of either laminar or turbulent flow. It has an accuracy of ±1% of the indicated flow.

7.11.2 General Considerations

The flux density (B) is related to the magnetic field intensity (H) by the relation

$$B = \mu H$$

where μ is called the *absolute permeability* and is expressed in Henry/metre. The absolute permeability of another material can be expressed relative to the permeability of free space by

$$\mu = \mu_r \, \mu_0$$

where μ_0 is the permeability of free space, its value is $4\pi \times 10^{-7}$ H/m, and it is a constant. μ_r is a dimensionless quantity called *relative permeability* and its value depends on the material of fluid.

Like other forms of flowmeter, the electromagnetic flowmeter requires a minimum length of straight pipe work upstream from the meter in order to guarantee the accuracy of measurement. No pipe bending or pipe fitting are usually allowed in five pipe diameter upstream.

7.11.3 Disadvantages

The electromagnetic flowmeter requires a minimum conductivity of 10 μmho cm^{-1}. Therefore, it is not suitable for measurement of flow of gases and liquid hydrocarbon. It is suitable for measurement of flow of slurries provided that the liquid phase has adequate conductivity. The instrument is expensive in terms of initial purchase cost. One of the reasons for high cost is the need for careful calibration of each instrument individually during manufacture as there are considerable variations in the properties of magnetic materials used.

7.11.4 Field Excitation

There are three possible ways of energising the magnetic field coils and they are as follows:

1. Direct current (dc).
2. Alternating current (ac) with 50 Hz.
3. Pulsating dc.

The three common difficulties faced by dc excitation are:

1. Polarisation effect.
2. Electrochemical effect.
3. Thermoelectric effect.

All these will influence the dc output voltage. These effects can be overcome by energising the field coils with an ac of 50 Hz. For many years ac excitation was most common. Signals can be easily amplified by high gain ac amplifier.

The major disadvantage of ac energised field coil is that the output voltage is subject to 50 Hz interference voltages generated by the transformer action in a loop consisting of the signal leads and the fluid path. The problem of ac and ac magnetic fields can be solved by energising the coils with pulsating dc or interrupted DC. It is direct current that is pulsed at a fixed interval. The pulsed magnetic field and the induced or output voltage of the flowmeter are shown in Fig. 7.26.

The output voltage is given by

$$e = e_2 - \frac{(e_1 + e_3)}{2} - e_4 - \frac{(e_3 + e_5)}{2}$$

Here the dc field is switched in a square wave fashion between some value and zero at frequency of 3 to 6 Hz. When the field is zero, any non-zero output from the flowmeter is considered to be an error. It is called a *zero error*. By storing the zero error and subtracting it from the output obtained when the field is next applied, the error can be removed. (The principle is similar to autozeroing of digital multimeter when the input is momentarily grounded and the non-zero output voltage charges a capacitor. This capacitor voltage can be utilised later to correct the output in the case of actual reading.) In practice, the zero error correction is done several times in a second.

7.11.5 *Electromagnetic Flowmeter with Metal Pipe*

For measurement of flow of liquid that has high conductivity, a metal pipe without insulating liner is used. For example, if the liquid is mercury, stainless-steel pipe will be used. Here the conductivity of

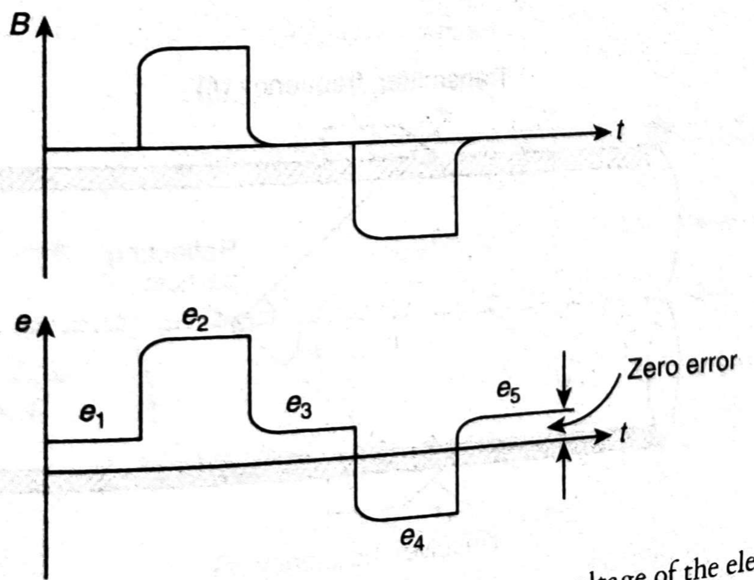

Figure 7.26 Magnetic flux produced by field coils and the output voltage of the electromagnetic flowmeter.

mercury is much higher than that of stainless steel. The electrode in this can be installed on the outside surface of the pipe.

7.12 Ultrasonic Flowmeter

The ultrasonic signal is short burst of sine waves when frequency is above the available range of frequency which is 20 kHz. The typical frequency of ultrasonic wave is 10 MHz. The ultrasonic measurement of volume flow rate is a non-invasive method. There are two types of ultrasonic flowmeters, namely, Doppler shift and transit time. Both the methods depend on transmitting and receiving of acoustic energy. The piezoelectric crystals are used for both the functions. In a transmitter electrical energy in the form of a short burst of high frequency voltage is applied to the crystal, causing it to vibrate. The crystal is in contact with the fluid and the vibration is communicated to the fluid and propagated through it. The vibration reaches the receiving crystal which produces an electric signal as an output.

7.12.1 *Doppler Shift Ultrasonic Flowmeter*

7.12.1.1 Principle of Operation

A fundamental requirement of this instrument is the presence of scattering elements within the flowing fluid. The Doppler meters will not work unless sufficient reflecting particles and/or air bubbles are present. This flowmeter usually employs clamp-on configuration as shown in Fig. 7.27.

The transmitter propagates ultrasonic wave of frequency ranging from 0.5 to 10 MHz in the fluid which is flowing with a uniform velocity v. The particles or bubbles moving with the same velocity will reflect some of the energy to the receiver. The reflecting elements cause a frequency shift between the transmitted and reflected ultrasonic energy. The frequency is measured and it is calibrated in terms of flow velocity.

The flow velocity (v) is given by

$$v = \frac{c(f_t - f_r)}{2 f_t \cos\theta}$$

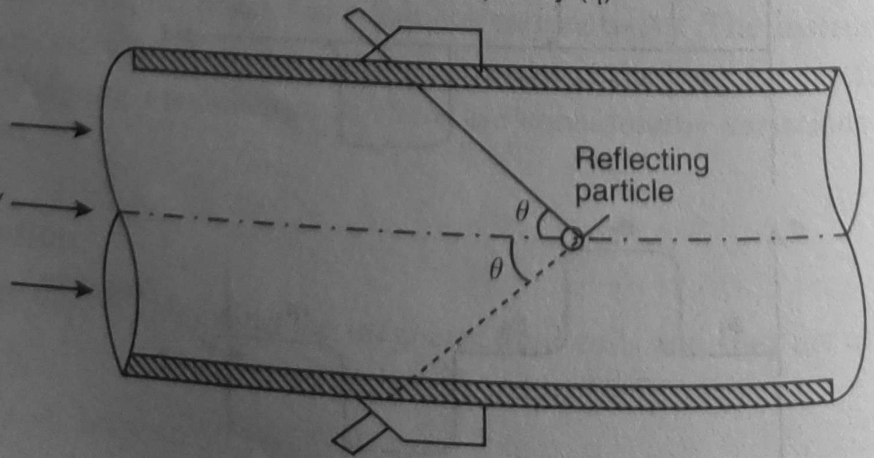

Figure 7.27 Doppler shift ultrasonic flowmeter.

where f_t is the frequency of the transmitted ultrasonic wave; f_r is the frequency of the received ultrasonic wave; c is the velocity of sound in the fluid being measured; θ is the angle of incidence which the reflected ultrasonic wave also makes with the axis of the flow in the pipe.

Both the transmitters and receivers are made with piezoelectric oscillator technology. The Doppler shift flowmeter is relatively inexpensive. The measurement accuracy depends on flow profile, the constancy of pipe wall thickness, the number and size of the reflecting particles, and the accuracy with which the velocity of sound in the fluid is known. The accurate measurement can only be achieved by carefully calibrating the instrument in each particular flow measurement application. Recently, the Doppler shift ultrasonic flowmeter has been developed with the transmitter and receiver flush in the inner surface of the pipe. In this, the problem of variable pipe thickness can be avoided. However we must note the following points while measuring the fluid velocity by shift flowmeter with clamp-on design:

1. Dependence of 'c' velocity of sound in fluid causes compensating changes in $\cos\theta$.
2. For such a design 'θ' is transducer wedge angle and 'c' is the propagation velocity of the wave in the wedge material.

7.12.2 *Transit Time Ultrasonic Flowmeter*

The transit time ultrasonic flowmeter is designed for measuring the volume flow rate in clean liquid. It consists of a pair of ultrasonic transducers mounted along an axis aligned at an angle θ with respect to the fluid flow axis. Transit time ultrasonic flowmeter is shown in Fig. 7.28. Each transducer consists of a transmitter–receiver pair. The transmitter emits ultrasonic energy which travels across to the receiver on the other side of the pipe. Fluid flowing through the pipe causes a time difference between the transit times of the beams traveling upstream and downstream, and measurement of the difference of time of travel gives the flow velocity. Typically the time difference is 100 ns in a total transit time of 100 μs.

7.12.3 *Methods of Measuring Time Shift*

There are three different methods of time shift and they are:

1. Direct measurement.
2. Conversion to phase change
3. Conversion to frequency change.

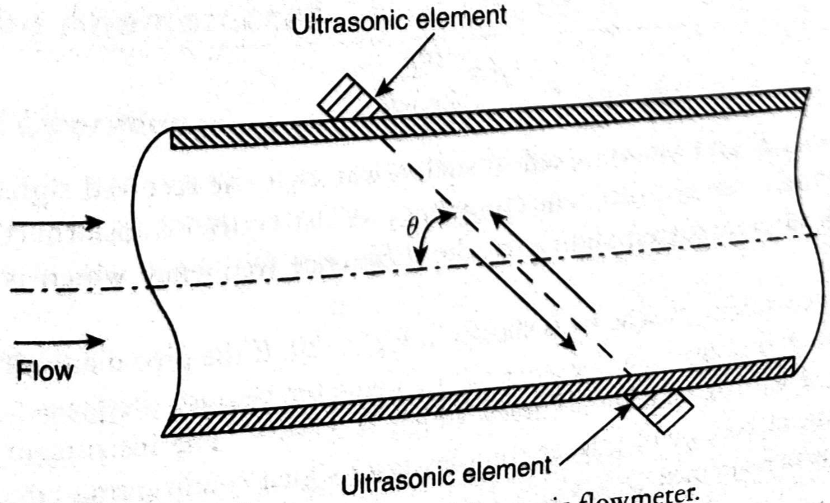

Figure 7.28 Transit time ultrasonic flowmeter.

The third method is attractive since it does not need to measure the speed of sound in the measured fluid. The method also multiplexes the transmitting and receiving functions so that same transducer can be used both as transmitter and receiver.

The forward and reverse transit time across the pipe t_f and t_r are given by

$$t_f = \frac{l}{c + v\cos\theta}$$

$$t_r = \frac{l}{c - v\cos\theta}$$

where c is the velocity of sound in the fluid; v is the velocity of the fluid; l is the distance between the ultrasonic transmitter and receiver; θ is the angle between the ultrasonic beam and axis of the fluid flow.

The time difference δt is given by

$$\delta t = t_r - t_f = \frac{2vl\cos\theta}{c^2 - v^2\cos^2\theta}$$

However, in actual practice the receipt of the pulse is used to trigger the transmission of next ultrasonic energy pulses. Thus, the frequencies of the forward and return pulse trains are given by

$$F_f = \frac{1}{t_f} = \frac{c + v\cos\theta}{l}$$

$$F_r = \frac{1}{t_r} = \frac{c - v\cos\theta}{l}$$

If the two frequency signals are now multiplied together, the resulting beat frequency is given by

$$\delta f = F_f - F_r = \frac{2v\cos\theta}{l}$$

Therefore

$$v = \frac{l\delta f}{2\cos\theta}$$

The two frequencies F_f and F_r are mixed in such a way that the received signal contains the beat frequencies that are their sum and difference frequency. A fast Fourier transform (FFT) analysis of this output will lead us to peak corresponding to the difference frequency which is a measure of flow velocity.

The transit time measurement scheme is shown in Fig. 7.29. If the pipe diameter is large, the transit time flowmeter is preferred over the Doppler shift flowmeter because the transit time is sufficiently large to be measured with reasonable accuracy which is ±0.5%. The instrument costs more than a Doppler shift flowmeter because of greater complexity of signal conditioning circuit needed to make accurate transit time measurement.

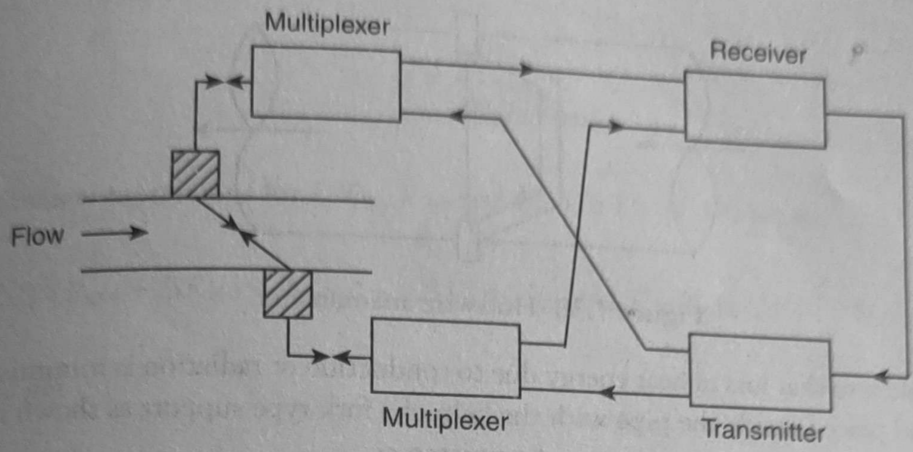

Figure 7.29 Transit time measurement system.

7.12.4 Advantages of Ultrasonic Flowmeter

The fluids flowing through the pipe is not necessarily conductive fluids. It is particularly useful for measuring the flow of corrosive fluids and slurries. A further advantage over other flowmeters is that the instrument is one which clamps on externally to existing pipe rather than being inserted as an integral part of the flow line. Unlike other flowmeters, the ultrasonic flowmeter can be installed without pipe cutting or breaking. Therefore, it has enormous cost advantage. Its clamp-on mode of operation has significant safety advantage in avoiding the possibility of personnel installing flowmeters coming into contact with hazardous fluid such as toxic, radioactive, flammable ones. Also any contamination of the fluid being measured can be easily avoided. Ultrasonic flowmeters have been used mainly for liquids but are now being used for gases too.

7.12.5 Disadvantages

Ultrasonic flowmeter is sensitive to velocity profile of the flow. It will give different readings for axisystemetic profiles of different shape but with identical average velocity. If we assume the meter co-efficient to be '1' for a uniform profile, this would drop to 0.75 for laminar flow and it will vary between 0.93 and 0.96 for turbulent flows.

7.13 Hot-Wire Anemometer

7.13.1 Principle of Operation

It is based on the principle of heat transfer by convection between the resistance wire and the flowing fluid. The wire is kept at a temperature much higher than the surrounding temperature and hence the name 'hot-wire'. It is capable of measuring average velocity and turbulent flow of the fluid.

7.13.2 Element Used

The hot-wire filament is usually a fine wire of platinum or tungsten.
The typical dimensions are:

1. Diameter: 5 μm–300 μm
2. Length: 1 mm–10 mm

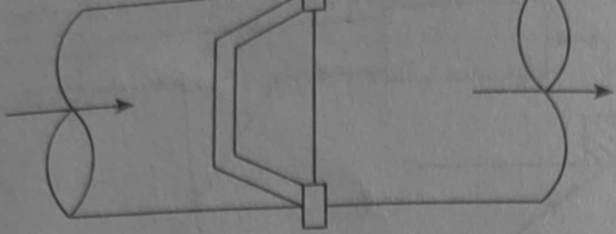

Figure 7.30 Hot-wire anemometer.

Care needs to be taken so that loss of heat energy due to conduction or radiation is minimised. The element is thus jacketed and placed inside the pipe with the help of a fork type support as shown in Fig. 7.30. At equilibrium, the heat balance equation can be written as

$$I^2 R_{WO} = U(v_0)A(T_{WO} - T_F) \tag{7.17}$$

that is the heat generated in wire is equal to the convective loss of heat from its surface. In Eq. (7.17), I is the current flowing through the resistance wire (in Ampere); R_{WO} is the resistance of the wire (in ohm); A is the heat transfer area [in (metre)^2]; T_{WO} is the temperature of the wire (°C); T_F is the temperature of the fluid (°C); $U(v_0)$ is the convective heat transfer coefficient (in $\text{W}\,\text{m}^{-2}\,°\text{C}^{-1}$). Mathematically

$$U(v_0) = a + b\sqrt{v}$$

where

$$a, b = f(d, k, \rho, \eta)$$

Here d is the sensor dimensions (in metre); k is the thermal conductivity of the fluid (in $\text{W}\,\text{m}^{-1}\,°\text{C}^{-1}$); ρ is the fluid density (in $\text{kg}\,\text{m}^{-3}$); η is the fluid viscosity (Pascal-sec).

7.13.3 Dynamic Characteristics

Taking into account the dynamics of the system, we can write

Rate of heat input – Rate of heat dissipation = Rate of rise of heat energy

We can write

$$I^2 R_W - U(v)A[T_W - T_F] = MC\frac{dT_W}{dt} \tag{7.18}$$

where M is the mass of heating element and C is the specific heat. At equilibrium or steady state we have Eq. (7.17).

For an incremental change we can write

$$I \rightarrow I_0 + \Delta I$$

$$T_W \rightarrow T_{WO} + \Delta T$$

$$R_W \rightarrow R_{WO} + \Delta R_W$$

$$U(v) \rightarrow U(v_0) + \sigma \Delta v$$

where

$$\sigma = \left.\frac{\partial U}{\partial v}\right|_{\substack{v=v_0 \\ T_W = T_{W0}}}$$

Substituting the new expressions for I, T_W, R_W, and $U(v)$ in Eq. (7.18) we get

$$(I_0 + \Delta I)^2 (R_{W0} + \Delta R_W) - \{U(v_0) + \sigma \Delta V\} A (T_{W0} + \Delta T - T_F) = MC \frac{d}{dt}(T_{W0} + \Delta T)$$

Simplifying we get

$$[2I_0 \Delta I R_{W0} + I_0^2 \Delta R_W] - U(v_0) A \Delta T - \sigma \Delta v A [T_{W0} - T_F] = MC \frac{d}{dt}(\Delta T)$$

Putting

$$\Delta T = \frac{\Delta R_W}{K_T}$$

we have

$$\Delta R_T = \frac{K_I}{1 + s\tau_v} \Delta I - \frac{K_v}{1 + s\tau_v} \Delta v$$

where

$$\tau_v = \frac{MC}{[U(v_0)A - I_0^2 K_T]}$$

$$K_v = \frac{K_T \sigma A [T_{W0} - T_F]}{[U(v_0)A - I_0^2 K_T]}$$

$$K_I = \frac{2K_T I_0 R_{W0}}{[U(v_0)A - I_0^2 K_T]}$$

The block diagram representation of hot-wire anemometer system is given in Fig. 7.31.

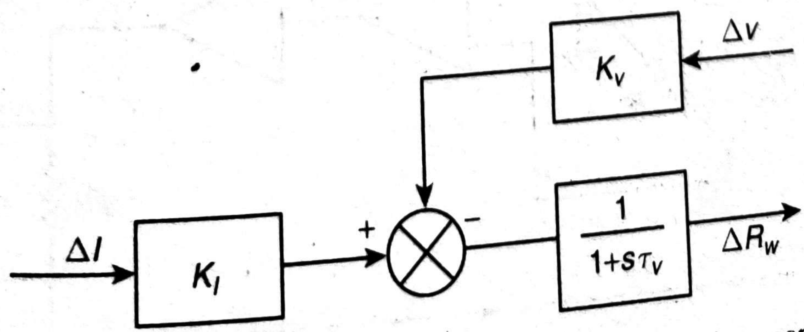

Figure 7.31 Block diagram representation of hot-wire anemometer.

7.13.4 Constant Current and Constant Temperature Type Hot-Wire Anemometer

There are two types of schematic arrangements for hot-wire anemometers:

1. Constant current type.
2. Constant temperature type (or constant R_W type).

A constant current type hot-wire anemometer with measuring bridge circuit is shown in Fig. 7.32. The current is kept fixed at a certain value. The resistances R_1, R_2, and R_3 are of the same order as R_W. The value of R_S is high. It is generally used to measure steady flow or low fluctuations in the velocity. It is not good for turbulent flow measurements.

A constant temperature type hot-wire anemometer is shown in Fig. 7.33. Here it keeps the resistance R_W constant by incorporating feedback. As the velocity increases, R_W decreases, thereby creating an unbalanced voltage. The increase of current brings back the resistance to the initial value. The feedback increases the bandwidth and therefore it is suitable for turbulent flow measurements also. The block diagram representation of constant temperature type is shown in Fig. 7.34.

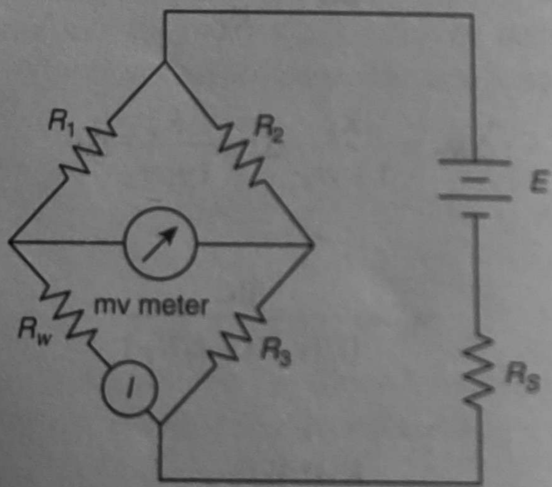

Figure 7.32 Circuit diagram of constant current type hot-wire anemometer.

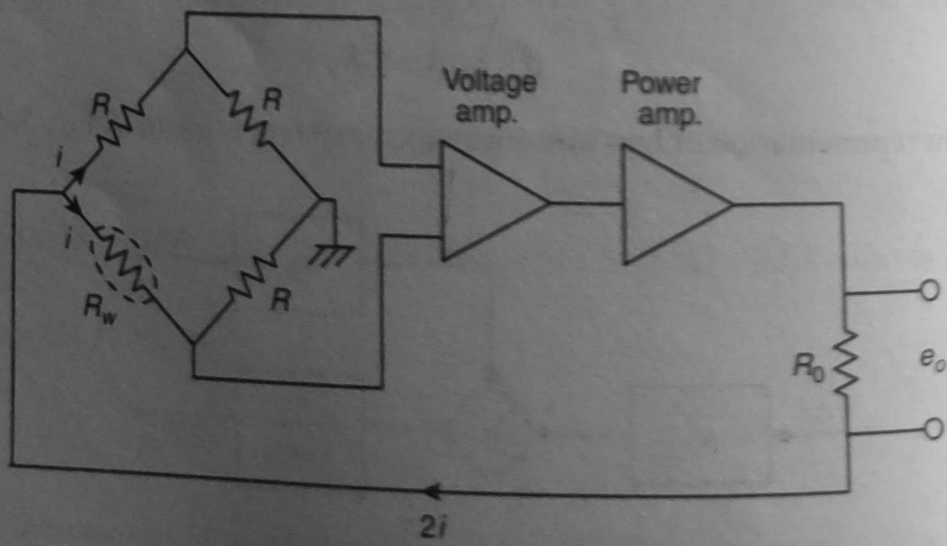

Figure 7.33 Current diagram of constant temperature type hot-wire anemometer.

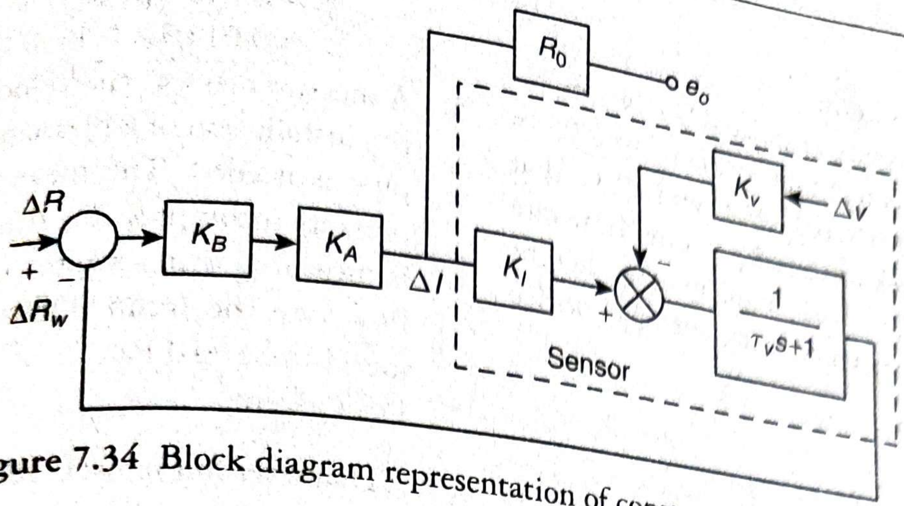

Figure 7.34 Block diagram representation of constant temperature type.

Analysis of the block diagram of Fig. 7.34 gives

$$\frac{\Delta e_o}{\Delta v} = \frac{K_v K_B K_A R_0}{\tau_v s + (1 + K_B K_A K_I)} = \frac{K'}{\tau'_v s + 1}$$

where

$$K' = \frac{K_v K_B K_A R_0}{1 + K_B K_A K_I}$$

$$\tau'_v = \frac{\tau_v}{1 + K_B K_A K_I}$$

Thus, we see that the time constant of the system reduces and the system becomes faster.

7.13.5 *Comparison of Constant Current and Constant Temperature Type Measurements*

Disadvantages of constant current hot-wire anemometer are as follows:

1. The current has to be kept large and with a sudden drop in fluid velocity it may lead to burn out of the wire.
2. For dynamic measurements separate compensating networks are required.

Constant temperature type removes the above problem with the introduction of a feedback loop. However, there are other problems such as instability and drift. For small velocity change noise has to be taken care of separately, otherwise the measurements would be erroneous.

7.8 Weir

Weir is an open-channel meter. If flow rate of a liquid in an open duct or channel such as a river or canal is required, a weir can be used. It is extensively used to get an estimate of flow of water in the canal or duct for irrigation purposes.

7.8.1 Principle of Operation of Weir

The flow rate over a weir is a function of the weir geometry and of the weir head. (The weir 'head' is defined as the vertical distance between the weir crest and the liquid surface in the undisturbed region of the upstream flow.)

The schematic of weir is shown in Fig. 7.18. The rectangular weir is shown in Fig. 7.19. Flow of liquid over the sharp edge of weir is shown in Fig. 7.20. In Figs. 7.19 and 7.20, H is the head on the weir and h is the distance below the free surface of water where V_2 exists.

Applying Bernoulli's equation at undisturbed region of upstream flow and at the crest (sharp edge) of weir, we get

$$H + \frac{V_1^2}{2g} = (H - h) + \frac{V_2^2}{2g} \tag{7.10}$$

where V_1 is the upstream flow of the liquid and V_2 is the flow at the crest of the weir. Therefore

$$V_2 = \sqrt{2g\left(h + \frac{V_1^2}{2g}\right)}$$

$$Q = \frac{A_2}{\sqrt{1-\left(\frac{A_2}{A_1}\right)^2}} \sqrt{\frac{2(P_1-P_2)}{\rho}}$$

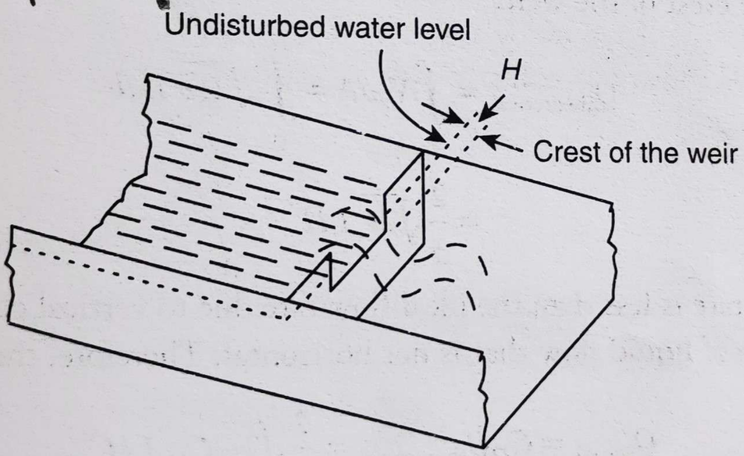

Figure 7.18 Schematic of a weir.

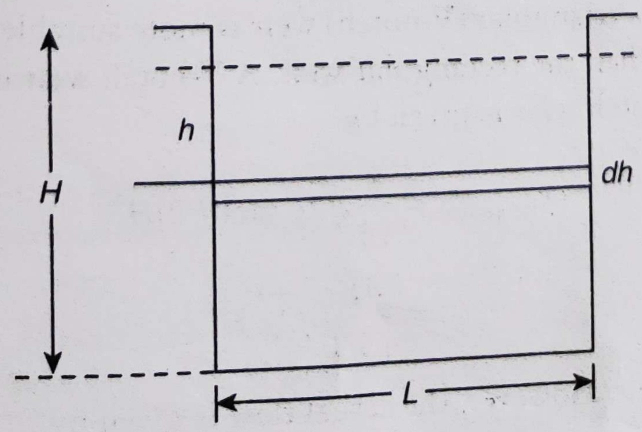

Figure 7.19 Rectangular weir.

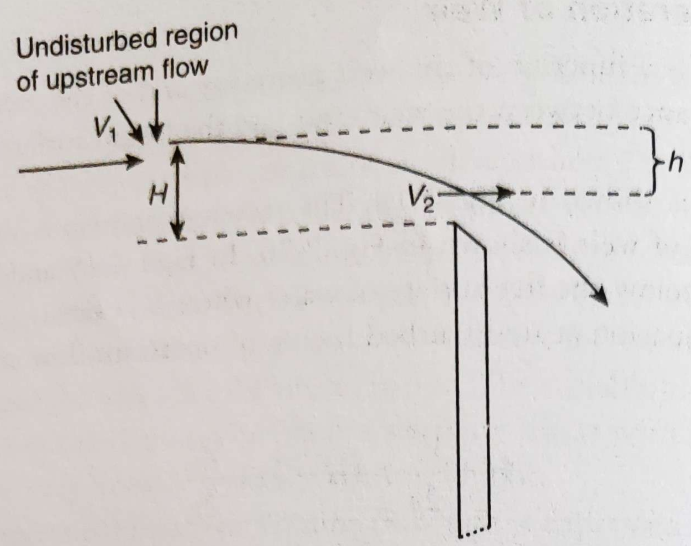

Figure 7.20 Flow over a sharp rectangular weir.

If V_1 is small compared to V_2, we get

$$V_2 \approx \sqrt{2gh} \tag{7.11}$$

The ideal flow rate over the weir is obtained by integrating the quantity $V_2 \times dA$ over the area (A) of the flow plane just above the crest of the weir:

$$Q_{\text{theoretical}} = \int_A V_2 dA = \int_0^H \sqrt{2gh}\, L dh$$

$$= \frac{2}{3}\sqrt{2g}\, LH^{3/2} \tag{7.12}$$

However, the actual flow rate is less than the ideal flow rate due to vertical contraction from the top, friction loss, and the presence of liquid flow that is not horizontal. Therefore, the actual flow rate is given by

$$Q_{\text{actual}} = C_D Q_{\text{theoretical}} = \frac{2}{3}\sqrt{2g}\, C_D\, LH^{3/2} \tag{7.13}$$

where C_D is the discharge co-efficient of rectangular weir and it lies between 0.62 and 0.75. The weir must be sharp for this co-efficient to be valid over a long period of time before the flowmeter is recalibrated.

When the flow rate is low, a triangular (V-notch) weir is more suitable for measurement of flow. This type of weir is more accurate than the rectangular weir. A V-notch weir is shown in Fig. 7.21.

The ideal flow rate for V-notch weir is given by

$$Q_{\text{theoretical}} = \frac{8}{15}\sqrt{2g}\, \tan(\theta/2) H^{5/2}$$

$$= \frac{4L}{15H}\sqrt{2g}\, H^{5/2} \tag{7.14}$$

where L is the width of the weir. However, the actual flow is given by

$$Q_{\text{actual}} = C_D \frac{4L}{15H}\sqrt{2g}\, H^{5/2} \tag{7.15}$$

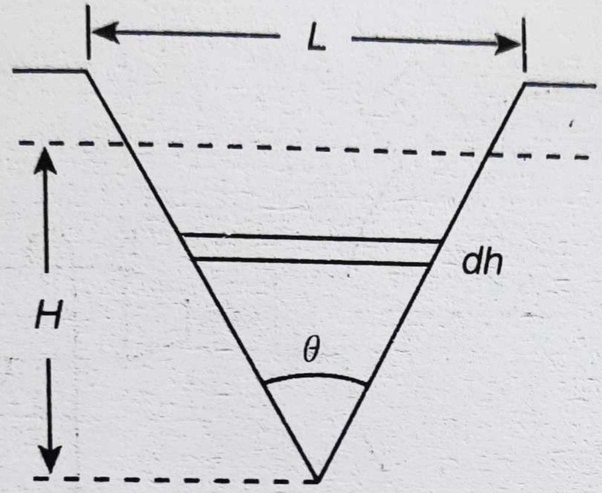

Figure 7.21 V-notch weir.

where C_D is the discharge co-efficient that lies between 0.58 and 0.60. Equations (7.13) and (7.15) indicate that the flow rate of the liquid depends on the head (H). The weir head is measured by a float activated displacement transducer and it is placed in the upstream side. It is usually used as an indicating instrument.